

Exam topics

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It is very important that you understand and can explain your code: the structure of your code, purpose of each function, logic behind your code, the algorithms you implemented and reasons for decisions you made. You must present the theory associated with relevant topics, a general overview is below.

1 General ML knowledge

- What is ML
- ML workflow
- Supervised and unsupervised learning
- Classification, regression, clustering, dimensionality reduction
- Binary and multiclass classification (OVR, OVO)
- Underfitting, overfitting
- Bias variance trade-off
- Loss function
- Hyperparameter tuning
- Regularization

2 Data preparation

- Issues with low-quality data
- Stages of data preparation (data exploration, data cleaning, feature selection, feature engineering, data transformation)
- Removal of irrelevant columns
- Handling of missing values (removal, imputation of numerical and categorical data)

- Reasons for missing data
- Handling outliers (keep, delete, impute, transform)
- Boxplots
- Skewed data, transformations
- Scaling (min-max, standard): algorithms that are sensitive to scaling
- Handling string data
- Bag of words
- One-hot encoding
- Handling correlating features

3 Validation methods

- Purpose of validation in ML workflow, iterative workflow
- Train-test
- Train-val-test
- Cross-validation (CV)
- Leave-one-out cross-validation (LOO CV)
- Use cases for each of the validation methods, advantages and disadvantages
- Use of test set only for final evaluation
- Use of validation set for evaluation and hyperparameter tuning
- Pre-processing and splitting data
- Issues with non-representative training data (stratification)
- Model generalization and overfitting

4 Performance metrics

- Regression: MSE, RMSE, MAE
- R-squared (coefficient of determination), SST, SSE, SSR
- Classification: accuracy, precision, recall, F1 score
- Classification errors

- Confusion matrix
- Precision-recall trade-off
- Precision-recall curve, ROC (Receiver operating characteristics) curve
- When to use which performance metric, what they represent

5 kNN

- Classification of new data points
- Hyperparameter k
- Even or odd k
- Distance metrics
- Importance of scaling

6 Linear regression

- Finding the best line
- Slope and intercept
- Least squares
- Design matrix, parameter vector, observation vector
- Simple linear regression
- Normal equation
- Regression performance metrics (MSE, MAE, RMSE, R-squared)
- Polynomial regression
- Ridge, Lasso and Elastic Net regression

7 Tree-based models

- Using for classification and regression
- Representation of a decision tree (binary directed acyclic graph, feature space)
- Gini impurity
- Interpretation of decision nodes and feature importance

- Pre-pruning (early-stopping)
- Ensembles (random forest, gradient-boosted decision trees)
- Randomness in random forest (bootstrapping/bagging, feature subsetting)
- Decision making in random forests
- Gradient-boosting, focus on misclassified instances
- Hyperparameters of decision trees, random forests and gradient-boosted decision trees
- Strengths and weaknesses of decision trees, random forests and gradient-boosted decision trees

8 Logistic regression

- Classification algorithm returning a continuous probability
- Logit
- Sigmoid function
- Impact of coefficient z on sigmoid function
- Log loss function
- Role of gradient descend in logistic regression
- Impact of learning rate
- Linearly non-separable data
- Multiple-features logistic regression

9 SVM

- Gutter, margin, support vectors
- Importance of scaling
- Impact of outliers on SVM
- Soft-margin SVM
- Slack variable ξ , margin violations
- Hyperparameter C
- SVM as an optimization problem for finding the hyperplane

- Non-linearly separable data, introduction of more dimensions
- Kernel functions
- Linear kernel, polynomial kernel, RBF (radial basis function) kernel

10 Neural networks

- Applications and relevance of NN
- Input layer, hidden layers, output layer
- Neurons, layers
- Forward propagation, weights, biases
- Role of activation functions for introduction of non-linearity
- Activation functions for hidden layers: ReLU, leaky ReLU, Tanh
- Activation functions for output layers: Softmax, Sigmoid, Linear
- Loss function
- Back-propagation, chain rule
- Role of optimizers
- SGD, epoch, batch, iteration
- Vanishing and exploding gradients
- Regularization with L1, L2, dropout and early stopping

11 Dimensionality reduction

- Difficulties of high dimensionality
- Approaches to dimensionality reduction (feature selection, regularization with L1/L2, PCA, t-SNE)
- Lossy and lossless compression
- PCA geometrically
- Orthogonality between principle components (PCs)
- Maximization of variance
- Principle components as linear combinations of features
- PCA algorithm

- Sample covariance matrix, variance, covariance, correlation
- Eigenvalues and eigenvectors, eigendecomposition, the characteristic equation
- Trace, captured variance
- Methods for choosing number of PCs (portion of captured variance, two, shoulder method)

12 Clustering

- Applications of clustering
- k -means
 - Centroids and clusters, iterative algorithm
 - Choice for initial centroids
 - Evaluation of clusters (inertia)
 - Choice for hyperparameter k
- DBSCAN (=Density-based spacial clustering of application with noise)
 - Density-based clustering
 - Core points, border points, noise
 - Hyperparameters n and ϵ and their effects
- Hierarchical (agglomerative) clustering
 - Bottom-up workflow
 - Merging based on closeness (distance metrics)
 - Dendrogram for finding an optimal number of cluster
 - Linkage (single, complete, average, Ward's)
 - WCSS as a metric
- Advantages, disadvantages and use cases for each algorithm